Toolkit Guide

Data Analysis Toolkit

Documentation

Author: Joel Payyappilly Elias Version: 1.0 Live App: LINK
Sample dataset: Titanic Dataset: LINK

Table of Contents

1.Overview

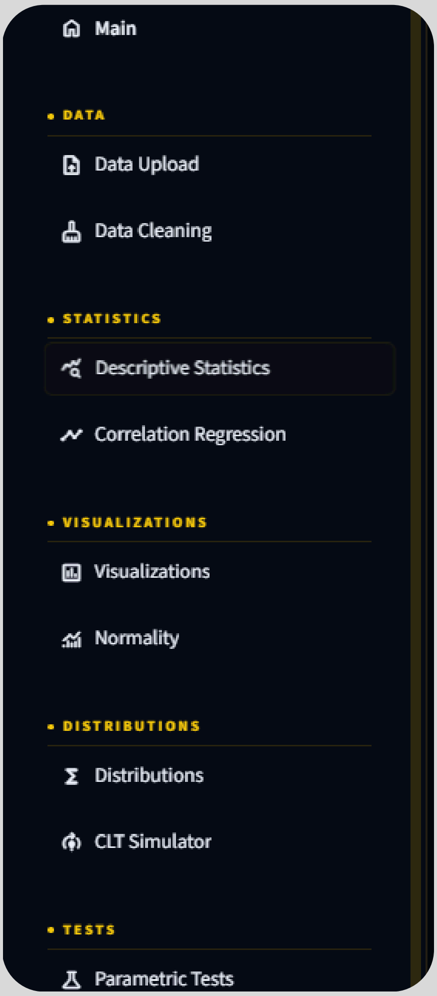
Data Analysis Toolkit is a statistical analysis platform built with Python and Streamlit.

What you can do:

- Upload a CSV or Excel dataset (up to 100,000 rows)
- Clean missing values and remove duplicates
- Explore your data with different tests
- Export results and plots as a PDF report

2.Getting Started

The sidebar on the left contains all pages grouped by category. Click any page name to navigate directly to it. The sidebar can be collapsed using the “<<” button at the top.

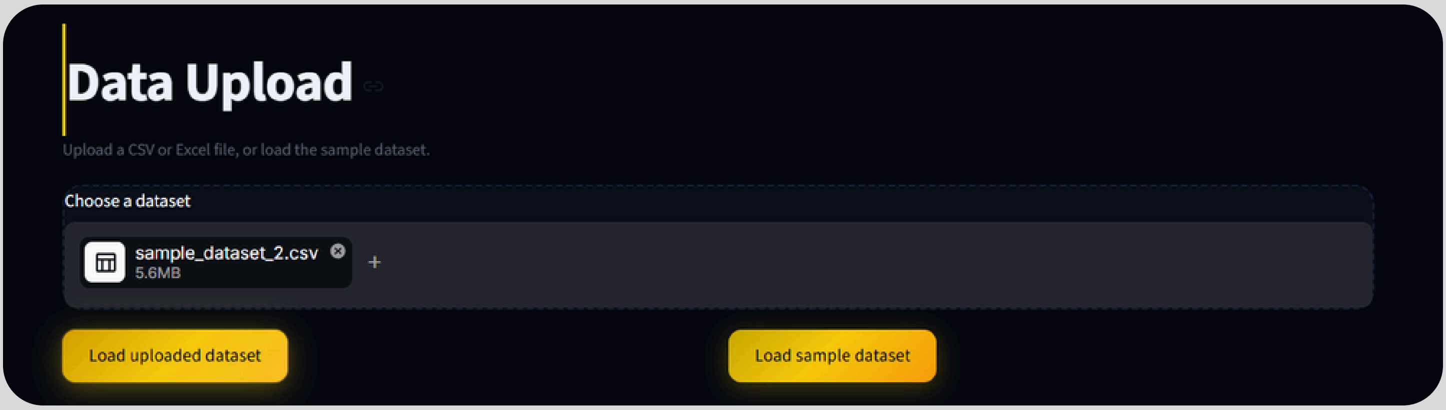


Recommended workflow:
Data Upload → Data Cleaning → Descriptive Statistics

→ Visualizations → Normality → Tests → Report

You must upload or load a dataset before using any analysis page. If you navigate to an analysis page without a dataset, you will see a prompt to go to “Data Upload first”

3. Data Upload



- Upload your own file - CSV, XLSX, or XLS up to 100,000 rows/200MB
- Load the sample dataset - A built-in dataset.
- Displays a dataset overview (row count, column count, missing value count, duplicate count)
- Shows detected column types - numeric vs categorical
- Shows a missing value table with counts and percentages per column
- Shows a preview of the first 50 rows

4. Data Cleaning

Apply one of the four missing value strategies, optionally removes duplicate rows, and optionally converts text columns to numeric.

Please make sure the missing values are null and not represented by “?” or other means

5. Descriptive Statistics

Descriptive Statistics

Summaries, correlations, outliers, recommendations, and automatic insights.

Descriptive Statistics | Correlation Matrix | Outlier Detection | Recommendations | Full Analysis

All Columns

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Student_ID	50000	None	None	None	125000.5	14433.9011	100001	112500.75	125000.5	137500.25	150000
Major_Category	50000	5	STEM	15059	None	None	None	None	None	None	None
Year_of_Study	50000	5	Junior	11045	None	None	None	None	None	None	None
Pre_Semester_GPA	49978	None	None	None	3.1461	0.4788	1.183	2.834	3.21	3.521	3.998
Weekly_GenAI_Hours	49981	None	None	None	8.4272	8.2689	0	2.39	5.8	11.72	40
Primary_Use_Case	50000	5	Debugging/Troubleshooting	12295	None	None	None	None	None	None	None
Prompt_Engineering_Skill	50000	3	Beginner	18495	None	None	None	None	None	None	None
Tool_Diversity	49987	None	None	None	2.8001	1.188	1	2	3	4	5
Paid_Subscription	49981	2	False	28831	None	None	None	None	None	None	None
Traditional_Study_Hours	49952	None	None	None	11.2092	5.1561	1	7.56	11.18	14.71	35.86

Numeric Detail

	count	mean	std	min	25%	50%	75%	max	variance	skewness	kurtosis
Student_ID	50000	125000.5	14433.9011	100001	112500.75	125000.5	137500.25	150000	208337500	0	-1.2
Pre_Semester_GPA	49978	3.1461	0.4788	1.183	2.834	3.21	3.521	3.998	0.2293	-0.6021	-0.1045
Weekly_GenAI_Hours	49981	8.4272	8.2689	0	2.39	5.8	11.72	40	68.3746	1.6098	2.5565
Tool_Diversity	49987	2.8001	1.188	1	2	3	4	5	1.4115	0.167	-0.7438
Traditional_Study_Hours	49952	11.2092	5.1561	1	7.56	11.18	14.71	35.86	26.585	0.1306	-0.3042
Perceived_AI_Dependency	50000	3.5054	1.8208	1	2	3	5	10	3.3154	0.6554	0.2627
Anxiety_Level_During_Exams	50000	4.2708	2.1441	1	3	4	6	10	4.597	0.3617	-0.4079
Post_Semester_GPA	50000	3.3493	0.4957	1	3.0738	3.421	3.749	4	0.2457	-0.6751	-0.034
Skill_Retention_Score	50000	75.7981	13.2836	10.78	66.82	76	85.19	100	176.4016	-0.2152	-0.2166

Tab 1 - Descriptive Statistics

Tab 2 - Correlation Matrix

- Select a correlation method (Pearson, Spearman, or Kendall) and view the full correlation matrix as a table and as a heatmap.

Tab 3 - Outlier Detection

- Lower bound (Q1 − 1.5 × IQR), Upper bound (Q3 + 1.5 × IQR), Count of outliers, Percentage of outliers

Tab 4 - Recommendations

- Automatic insights: plain-language observations about the dataset (size, missing values, most skewed column, highest-variance column)
- Recommended tests: suggests which statistical tests are appropriate based on what column types are present

Tab 5- Full Analysis

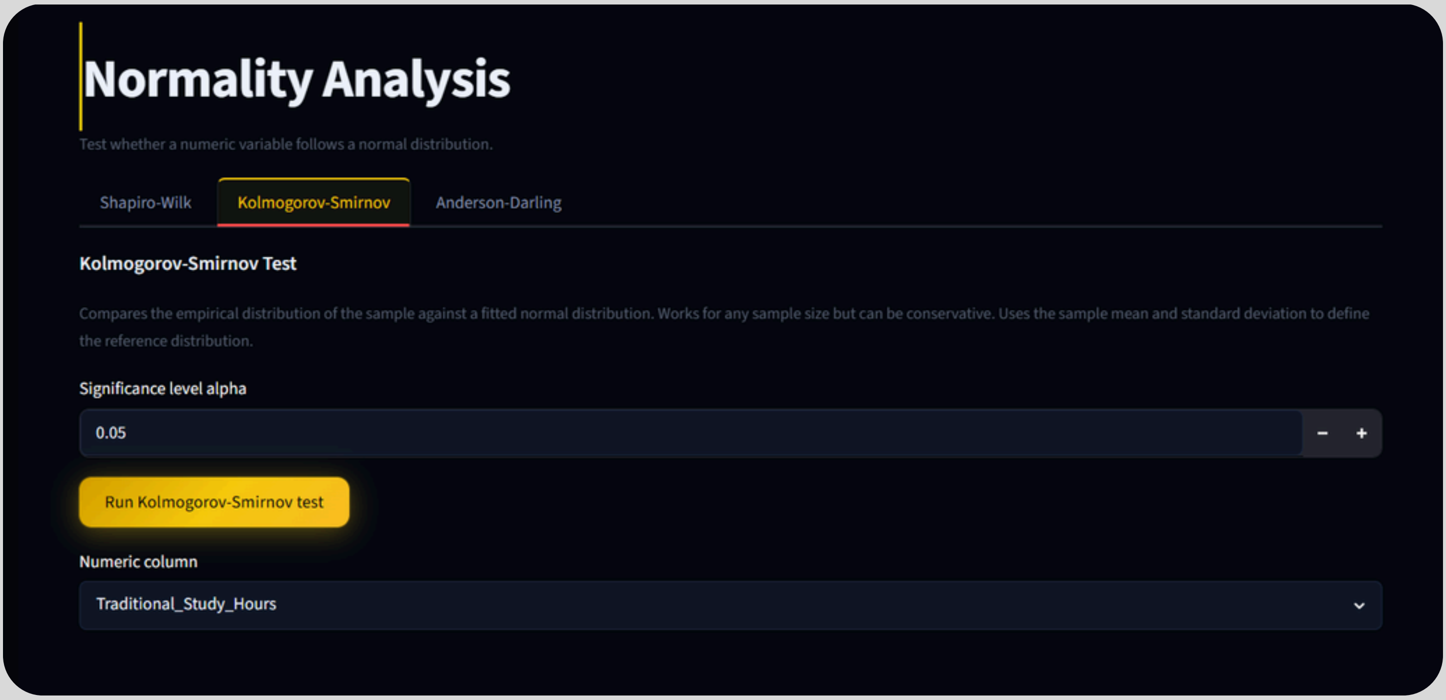
- Runs a combined view in one click: dataset overview, insights, descriptive statistics, normality preview, correlation matrix, and a histogram of the first numeric column.

6. Visualizations

Select a plot type from the dropdown and configure the inputs.
Every plot can be downloaded

Notes: All plots are saved to the session and can be included in the PDF report. The heatmap requires at least two numeric columns

7. Normality Analysis



Select a numeric column, then choose a test tab and run it independently.

<u>Shapiro-Wilk Test</u>

- Statistic: W (ranges from 0 to 1; values close to 1 suggest normality)
- Best for: Sample sizes up to ~5,000

- Note: For samples larger than 5,000, a random sub-sample of 5,000 is used
- Output: W statistic, p-value, decision, interpretation

Kolmogorov-Smirnov Test

- Statistic: D (maximum distance between the empirical and theoretical CDF)
- Best for: Any sample size
- Output: D statistic, p-value, decision, interpretation

Note: Uses the sample's own mean and standard deviation to define the reference normal distribution. P-values may be conservative.

Anderson-Darling Test

- Statistic: A^2 (gives more weight to the tails than KS)
- Best for: Any sample size. generally, more powerful than KS for normality
- Output: A^2 statistic, critical value at 5%, decision, full critical value table

Note: Does not produce a p-value. The decision is made by comparing A^2 against critical values at standard significance levels (15%, 10%, 5%, 2.5%, 1%).

All three tests display a histogram with fitted normal curve and a Q-Q plot after running. Results are logged to the analysis log for inclusion in the PDF report.

8. Distributions

Tab 1 - Normal Distribution

Enter a mean and standard deviation. Choose between PDF (probability density function) and CDF (cumulative distribution function). The plot and a data table of 400 points are displayed.

Tab 2 - Binomial Distribution

Enter the number of trials (n) and probability of success (p). Choose between PMF (probability mass function) and CDF. The full distribution table is shown alongside the plot.

Tab 3 - Poisson Distribution

Enter lambda (λ, the average rate). Choose between PMF and CDF.

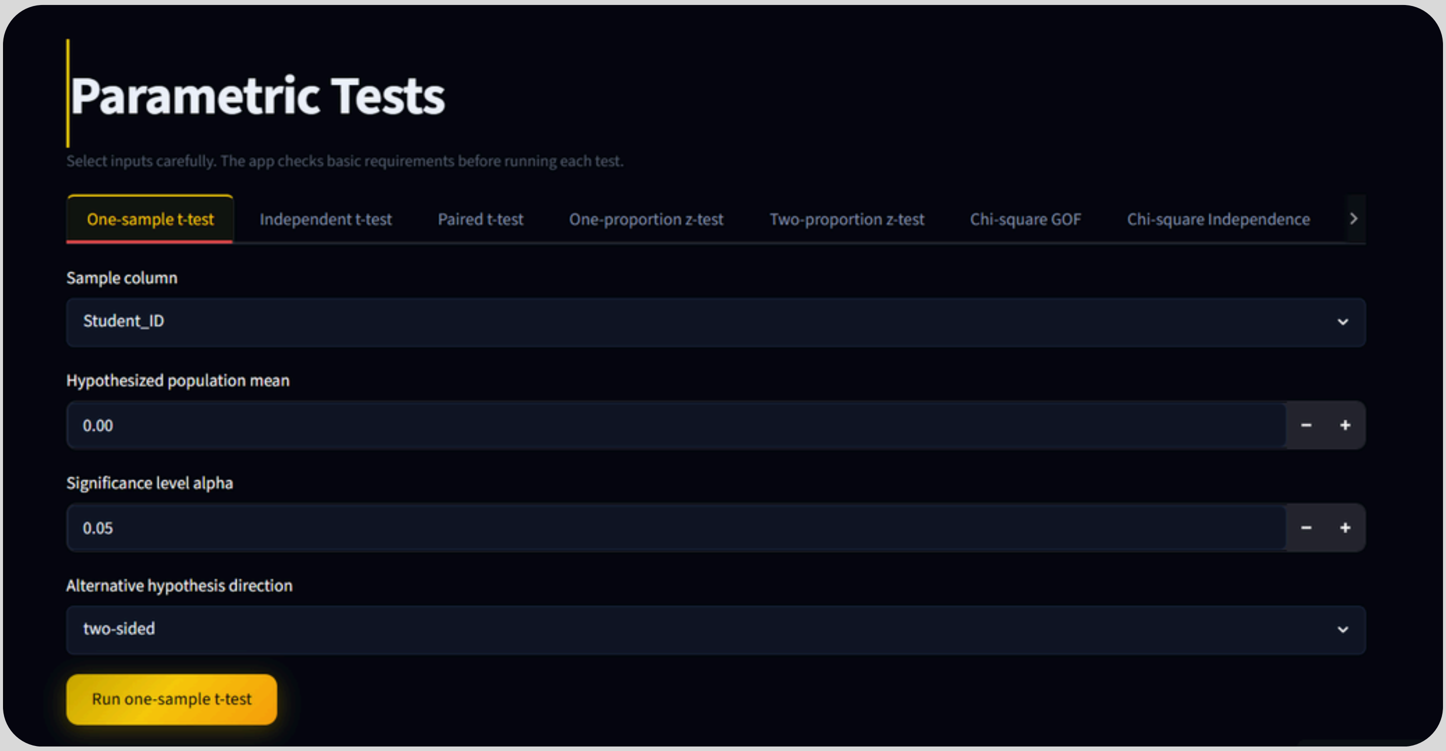
Tab 4 - Distribution Fitting

Select a numeric column from your dataset. The toolkit fits a normal distribution to the data using maximum likelihood estimation and overlays the fitted curve on a histogram.

Results include:

- Fitted mean and standard deviation
- Kolmogorov-Smirnov test statistic and p-value for goodness of fit
- A warning if the KS test suggests the normal fit is poor ($p < 0.05$)

9. Parametric Tests



Nine tests available as tabs. Each test includes configurable alpha (significance level) and alternative hypothesis direction where applicable.

One-sample t-test: Tests whether the sample mean equals a hypothesized population mean.

- Inputs: Numeric column, hypothesized mean, alpha, alternative
- Output: t statistic, p-value, degrees of freedom, decision

Independent t-test (Welch's): Tests whether two independent group means are equal.

- Inputs: Numeric outcome, categorical group column, select two groups, equal variance assumption, alpha, alternative
- Output: t statistic, p-value, Welch's degrees of freedom, decision
- Plot: Boxplot of both groups

Paired t-test: Tests whether the mean difference between two related measurements is zero.

- Inputs: Two numeric columns (before/after or matched pairs), alpha, alternative
- Output: t statistic, p-value, degrees of freedom, decision
- Plot: Histogram of paired differences

One-proportion z-test: Tests whether an observed proportion equals a hypothesized population proportion.

- Inputs: Number of successes, total observations, hypothesized proportion, alpha, alternative
- Output: z statistic, p-value, decision

Two-proportion z-test: Tests whether two independent proportions are equal.

- Inputs: Successes and observations for Group A and Group B, alpha, alternative
- Output: z statistic, p-value, decision

Chi-square Goodness-of-Fit: Tests whether observed category frequencies match expected frequencies.

- Inputs: Categorical column (counts are auto-computed), option for equal expected or manual expected counts, alpha
- Output: Chi-square statistic, p-value, degrees of freedom, decision
- Plot: Observed vs expected bar chart

Chi-square Independence: Tests whether two categorical variables are statistically associated.

- Inputs: Two categorical columns, alpha
- Output: Chi-square statistic, p-value, degrees of freedom, decision, expected counts table

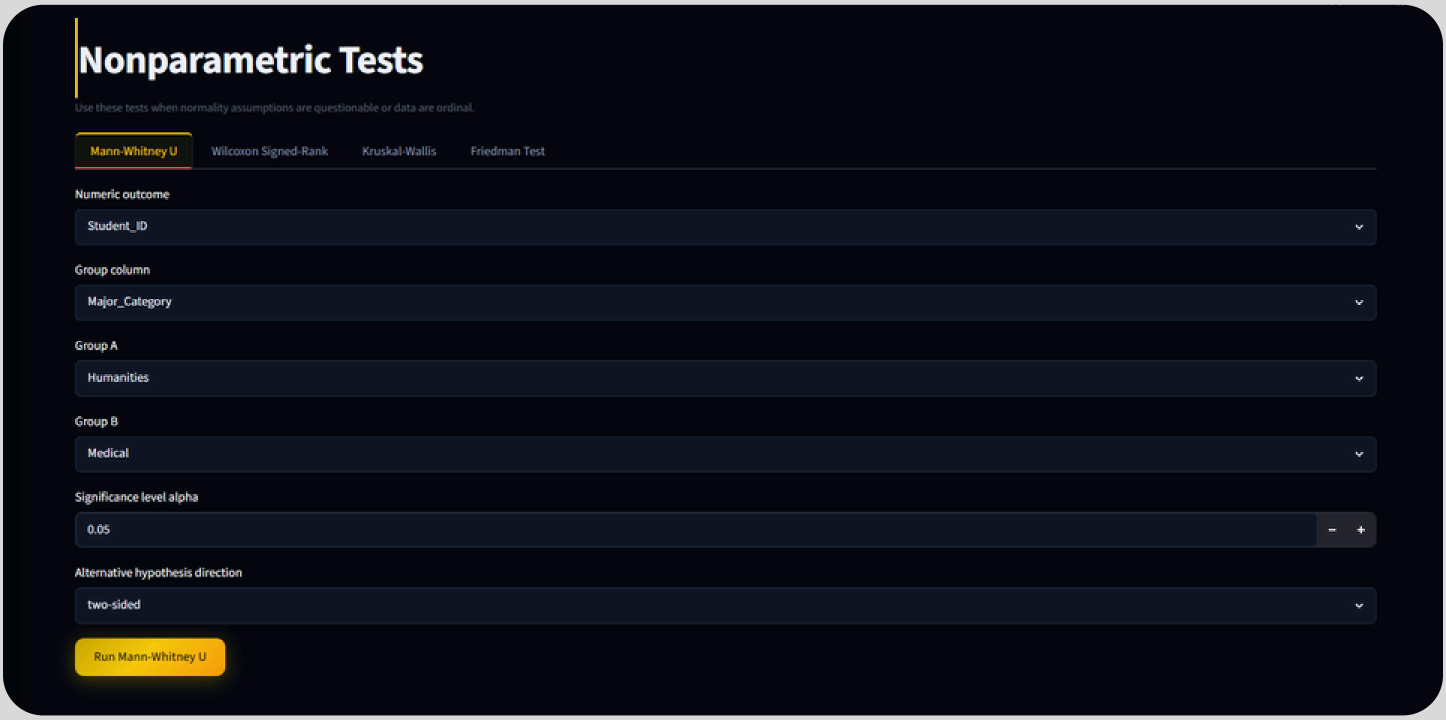
Pearson Correlation Significance: Tests whether the linear correlation between two numeric variables is significantly different from zero.

- Inputs: Two numeric columns, alpha
- Output: r statistic, p-value, degrees of freedom, decision
- Plot: Scatter plot

Regression Coefficient Significance: Tests whether the slope of a simple linear regression model is significantly different from zero.

- Inputs: Outcome variable, predictor variable, alpha
- Output: t statistic, p-value, coefficient, R^2, decision, full model summary
- Plot: Regression line overlaid on scatter plot

10. Nonparametric Tests



Mann-Whitney U Test: Nonparametric alternative to the independent t-test. Compares two independent groups.

- Inputs: Numeric outcome, categorical group column, two groups, alpha, alternative
- Output: U statistic, p-value, decision
- Plot: Boxplot of both groups

Wilcoxon Signed-Rank Test: Nonparametric alternative to the paired t-test. Compares two related measurements.

- Inputs: Two numeric columns, alpha, alternative
- Output: W statistic, p-value, decision
- Plot: Histogram of paired differences

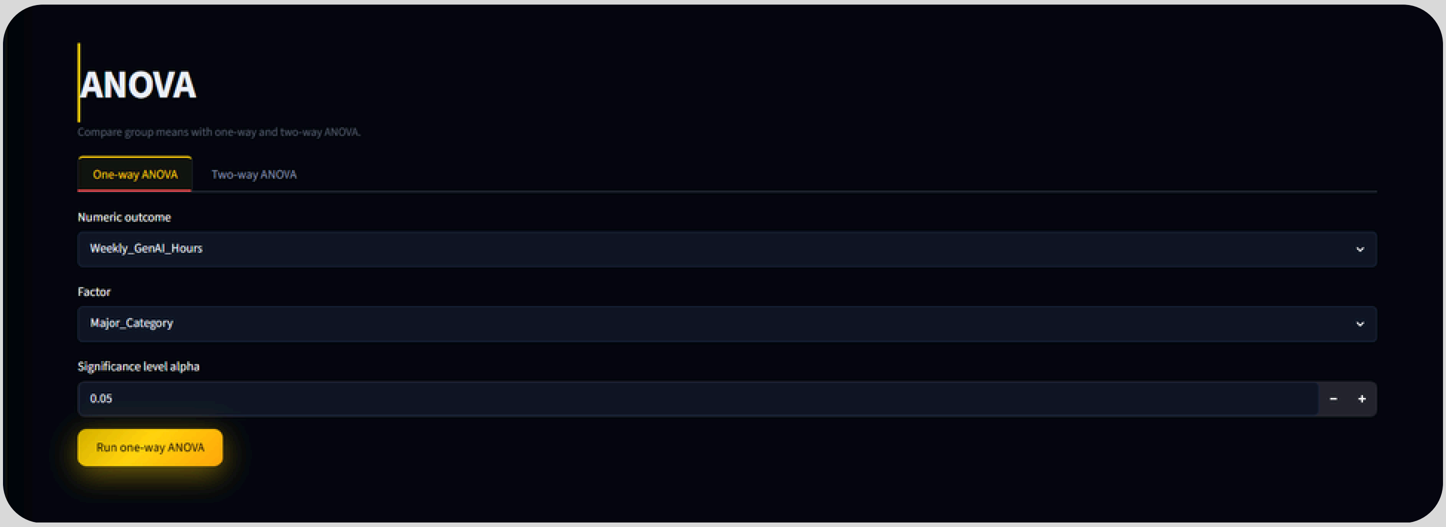
Kruskal-Wallis Test: Nonparametric alternative to one-way ANOVA. Compares three or more independent groups.

- Inputs: Numeric outcome, categorical group column, alpha
- Output: H statistic, p-value, decision
- Plot: Boxplot of all groups

Friedman Test : Nonparametric alternative to repeated-measures ANOVA.

- Inputs: Three or more numeric columns, alpha
- Output: Chi-square statistic, p-value, decision
- Plot: Line chart across all measurement columns

11. ANOVA



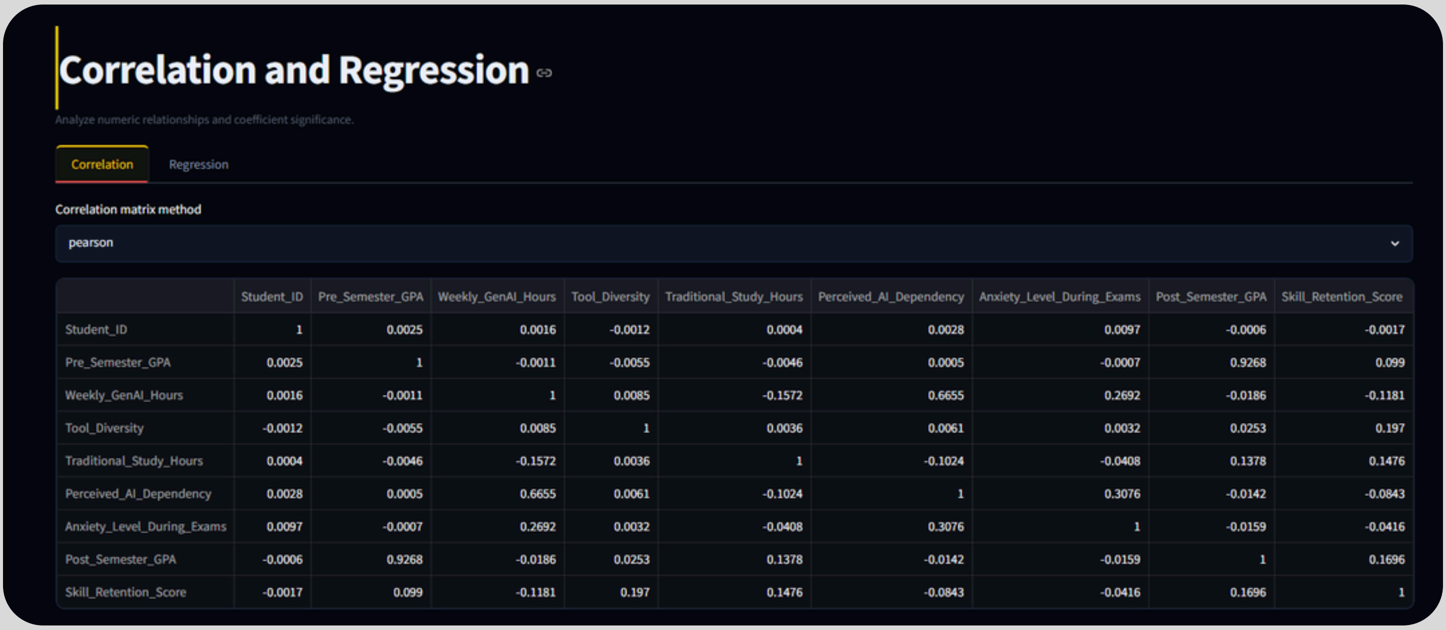
One-way ANOVA: Tests whether the means of a numeric outcome are equal across three or more groups defined by a single categorical factor.

- Inputs: Numeric outcome, categorical factor, alpha
- Output: F statistic, p-value, degrees of freedom (between, within), decision
- Plot: Boxplot grouped by factor

Two-way ANOVA: Tests the effect of two categorical factors and their interaction on a numeric outcome.

- Inputs: Numeric outcome, Factor A, Factor B, alpha
- Output: ANOVA table with rows for Factor A, Factor B, interaction, and residuals - each with F statistic, p-value, and decision
- Plots: Boxplot of outcome by Factor A; interaction plot showing Factor A × Factor B means

12. Correlation and Regression



	Student_ID	Pre_Semester_GPA	Weekly_GenAI_Hours	Tool_Diversity	Traditional_Study_Hours	Perceived_AI_Dependency	Anxiety_Level_During_Exams	Post_Semester_GPA	Skill_Retention_Score
Student_ID	1	0.0025	0.0016	-0.0012	0.0004	0.0028	0.0097	-0.0006	-0.0017
Pre_Semester_GPA	0.0025	1	-0.0011	-0.0055	-0.0046	0.0005	-0.0007	0.9268	0.099
Weekly_GenAI_Hours	0.0016	-0.0011	1	0.0085	-0.1572	0.6655	0.2692	-0.0186	-0.1181
Tool_Diversity	-0.0012	-0.0055	0.0085	1	0.0036	0.0061	0.0032	0.0253	0.197
Traditional_Study_Hours	0.0004	-0.0046	-0.1572	0.0036	1	-0.1024	-0.0408	0.1378	0.1476
Perceived_AI_Dependency	0.0028	0.0005	0.6655	0.0061	-0.1024	1	0.3076	-0.0142	-0.0843
Anxiety_Level_During_Exams	0.0097	-0.0007	0.2692	0.0032	-0.0408	0.3076	1	-0.0159	-0.0416
Post_Semester_GPA	-0.0006	0.9268	-0.0186	0.0253	0.1378	-0.0142	-0.0159	1	0.1696
Skill_Retention_Score	-0.0017	0.099	-0.1181	0.197	0.1476	-0.0843	-0.0416	0.1696	1

Correlation tab

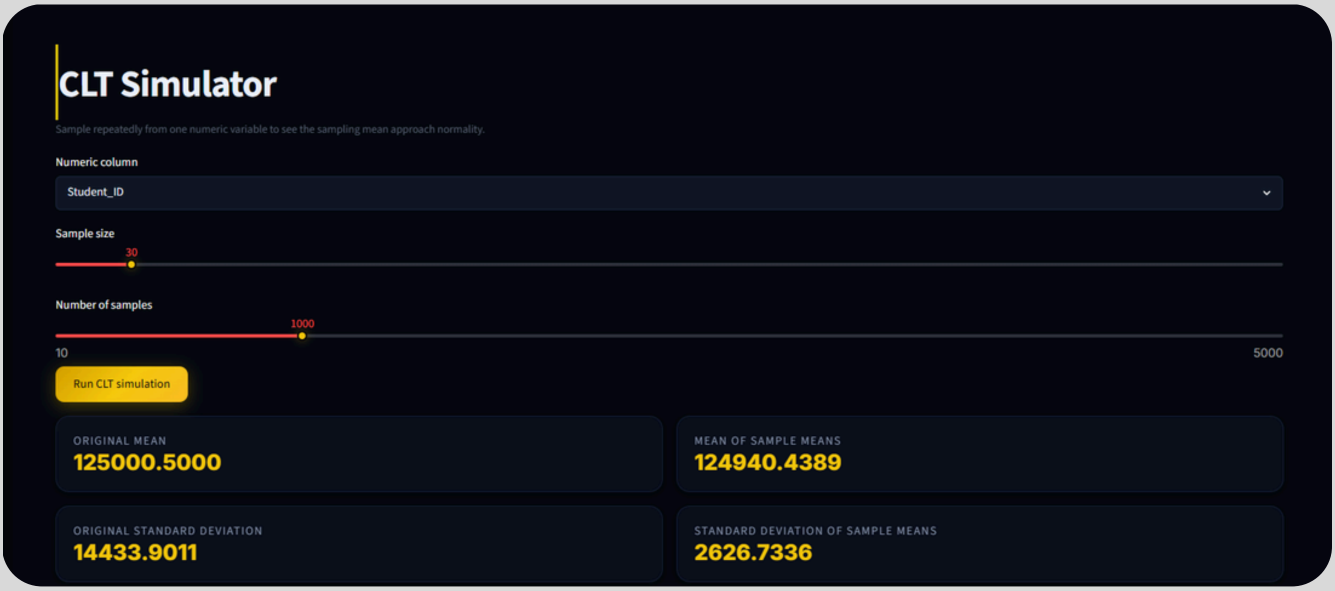
- Displays the full correlation matrix for all numeric columns using the selected method (Pearson, Spearman, or Kendall)
- Heatmap with red-to-yellow color scale
- Pearson correlation significance test for any two selected columns

Regression tab

Fits a simple linear regression model (one predictor, one outcome).

- Inputs: Outcome variable, predictor variable, alpha
- Output: Regression coefficient, t statistic, p-value, R^2, full statsmodels model summary, decision
- Plot: Scatter plot with regression line

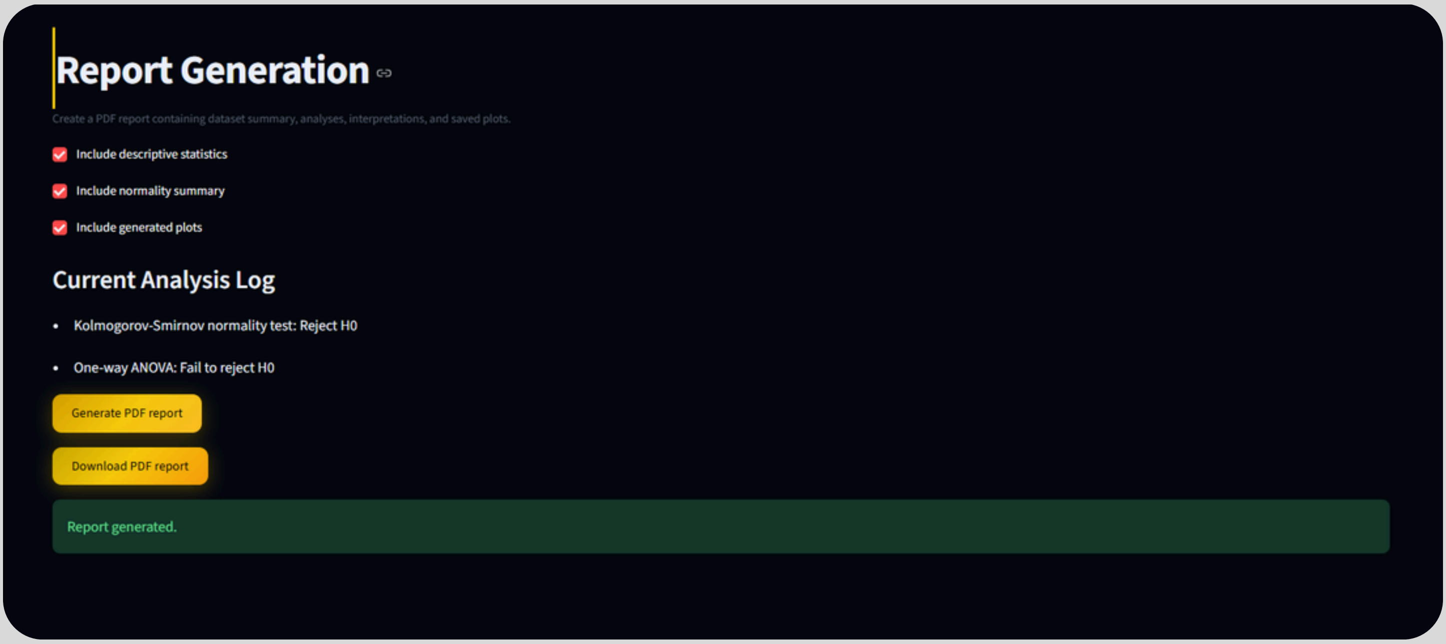
13. CLT Simulator



Demonstrates the Central Limit Theorem by repeatedly sampling from a real variable in your dataset and plotting the distribution of sample means.

As sample size increases, the sampling distribution becomes more bell-shaped regardless of the shape of the original distribution - this is the Central Limit Theorem in action.

14. Report Generation



What gets included

Section	Included when
Dataset summary	Always
Descriptive statistics table	Checkbox enabled
Normality summary table	Checkbox enabled
Analysis log	Any tests have been run during the session
Plots	Checkbox enabled and plots have been downloaded during session

Every time you run a test on any page and it succeeds, the result is added to the analysis log. The Report page shows the log before generating so you can review what will be included.

Notes: The PDF is generated using ReportLab and is compatible with all PDF viewers. Plots are embedded as PNG images at 150 DPI. The report is generated fresh each time. re-run tests or download plots again if the session was reset

15. Supported File Formats

Format	Extension	Notes
CSV	.csv	Standard comma-separated values
Excel	.xlsx	Excel 2007 and later
Excel (legacy)	.xls	Excel 97–2003 format

Maximum file size: 200 MB per file Maximum rows: 100,000 rows

16. Limitations

Limitation	Detail
Session-based	Data and results exist only for the current browser session. Closing the tab resets everything.
Single dataset	Only one dataset can be active at a time. Loading a new dataset replaces the current one.
Simple regression only	The regression module supports one predictor and one outcome. Multiple regression is not available.
No time series	Date/time columns are not specially handled.
Shapiro-Wilk sample cap	For datasets with more than 5,000 rows, Shapiro-Wilk uses a random sub-sample of 5,000.
Anderson-Darling p-value	The Anderson-Darling test does not return a p-value by design. Decisions are based on critical values only.
PDF plots	Only plots that were downloaded during the session (via the Download button) are included in the PDF report.